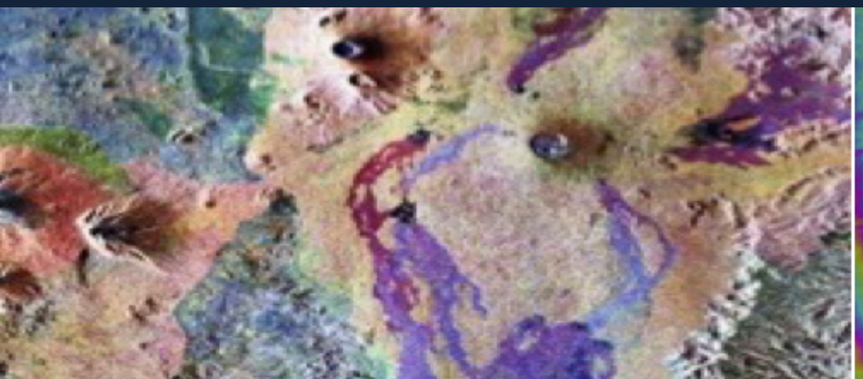
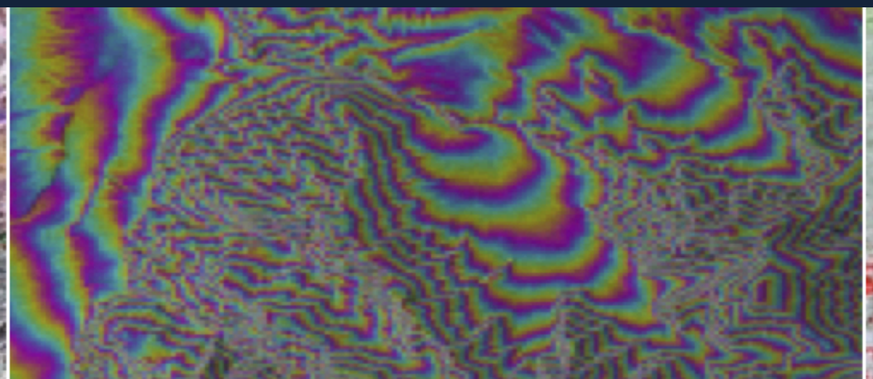
SYNTHETIC APERTURE RADAR INTERFEROMETRY · LECTURE 3

# InSAR Theory

AKSHAY RAJ MANOCHA







THIS LECTURE

# Outline

**01** **From SAR to InSAR**
SAR images, the electromagnetic spectrum, backscatter, and how two images form an interferogram

**02** **Interferometry for topography**
Baselines, the critical baseline, topographic phase and DEM missions

**03** **Interferometry for surface deformation**
Deformation phase, the Landers earthquake, sensitivity and phase unwrapping

**04** **Phase contributions and error sources**
Orbits, DEMs, the ionosphere and the troposphere

**05** **Correlation and wavelength**
Interferogram quality, decorrelation, and X-, C-, L- and P-band compared

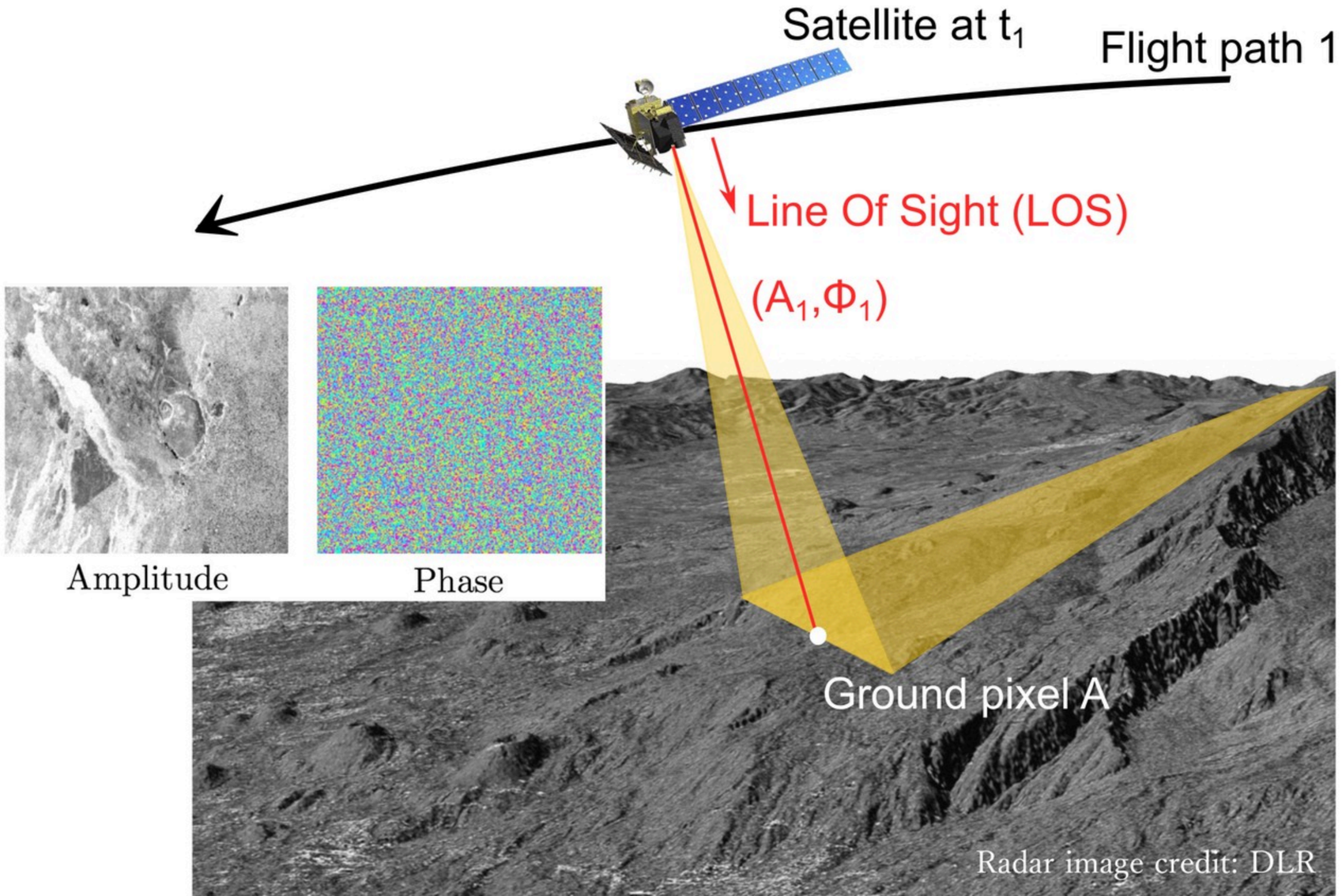
01

PART 1

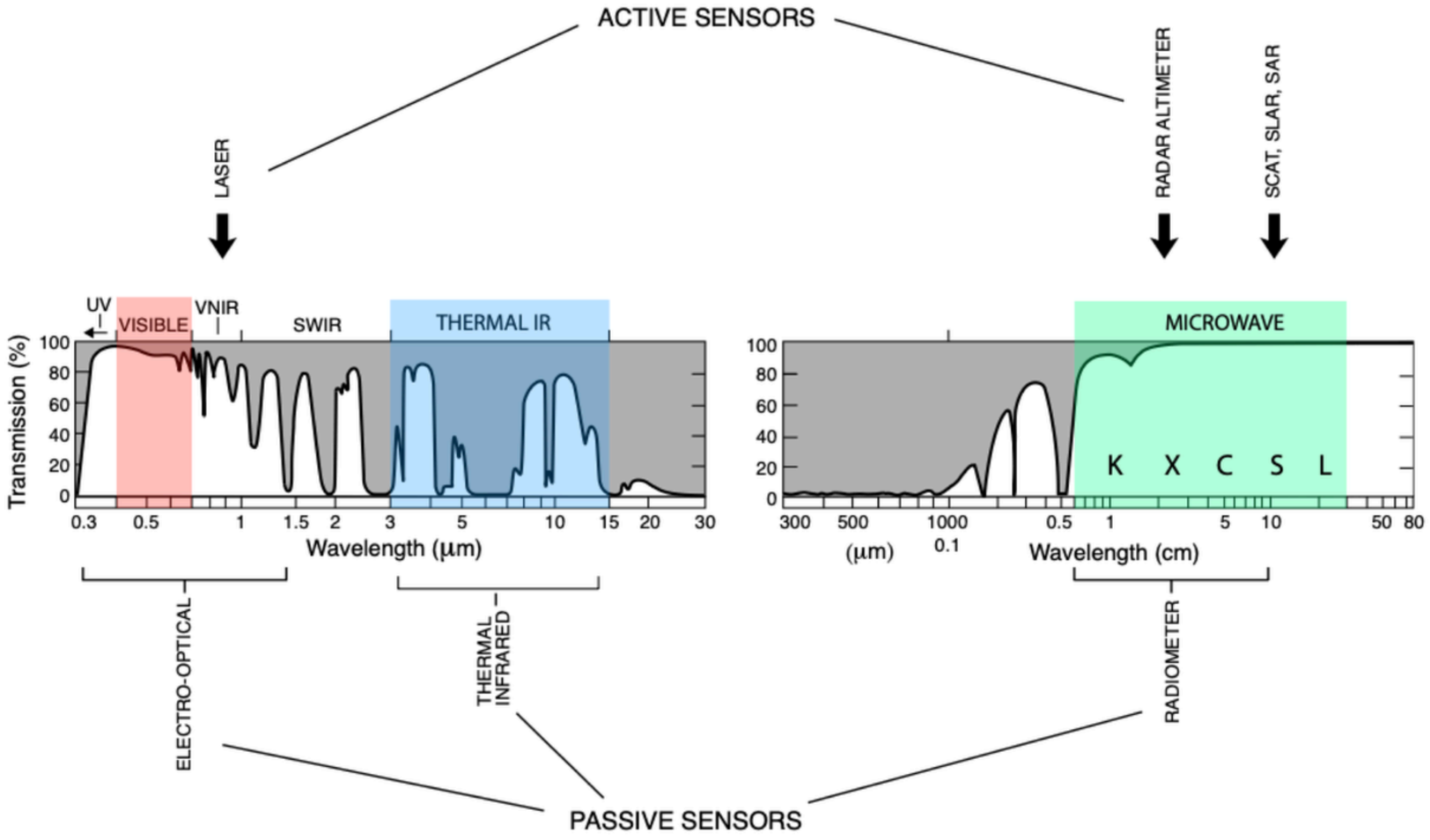
# From SAR to InSAR

SAR images, the electromagnetic spectrum, backscatter, and how two images form an interferogram

# Synthetic Aperture Radar (SAR)



# Electromagnetic Spectrum



# What's in a SAR image - backscatter

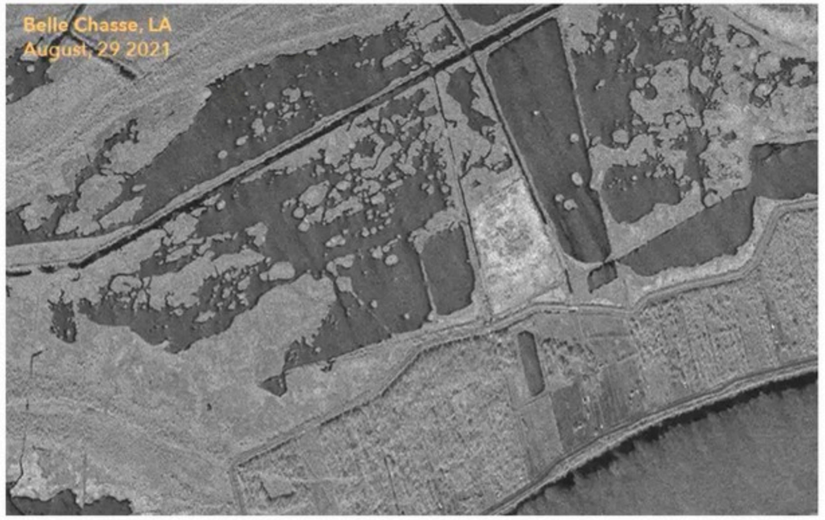
Day 1– normal conditions




Day 3 – flooded and possibly windy



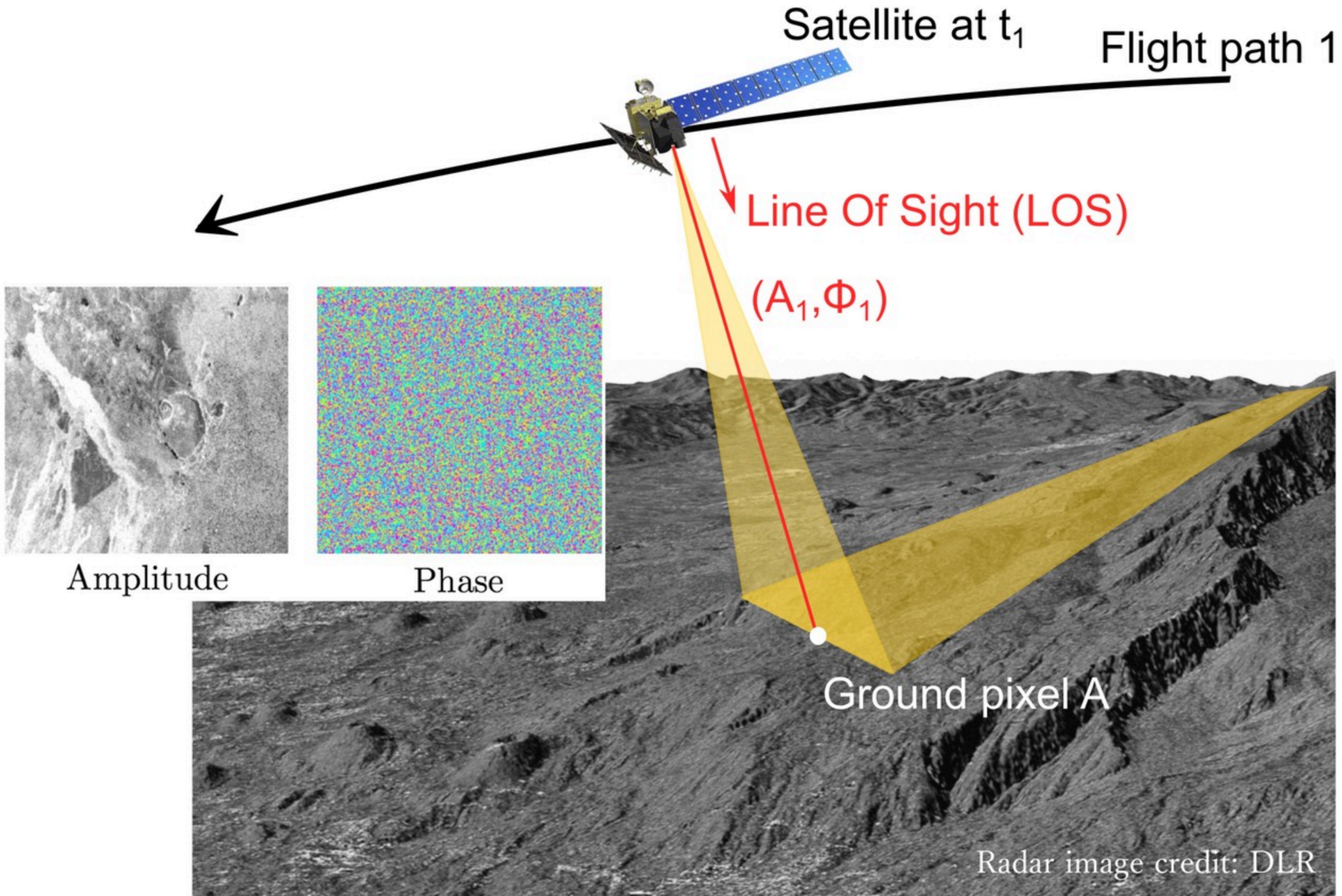

Day 2 – windy (high reflectivity on water)



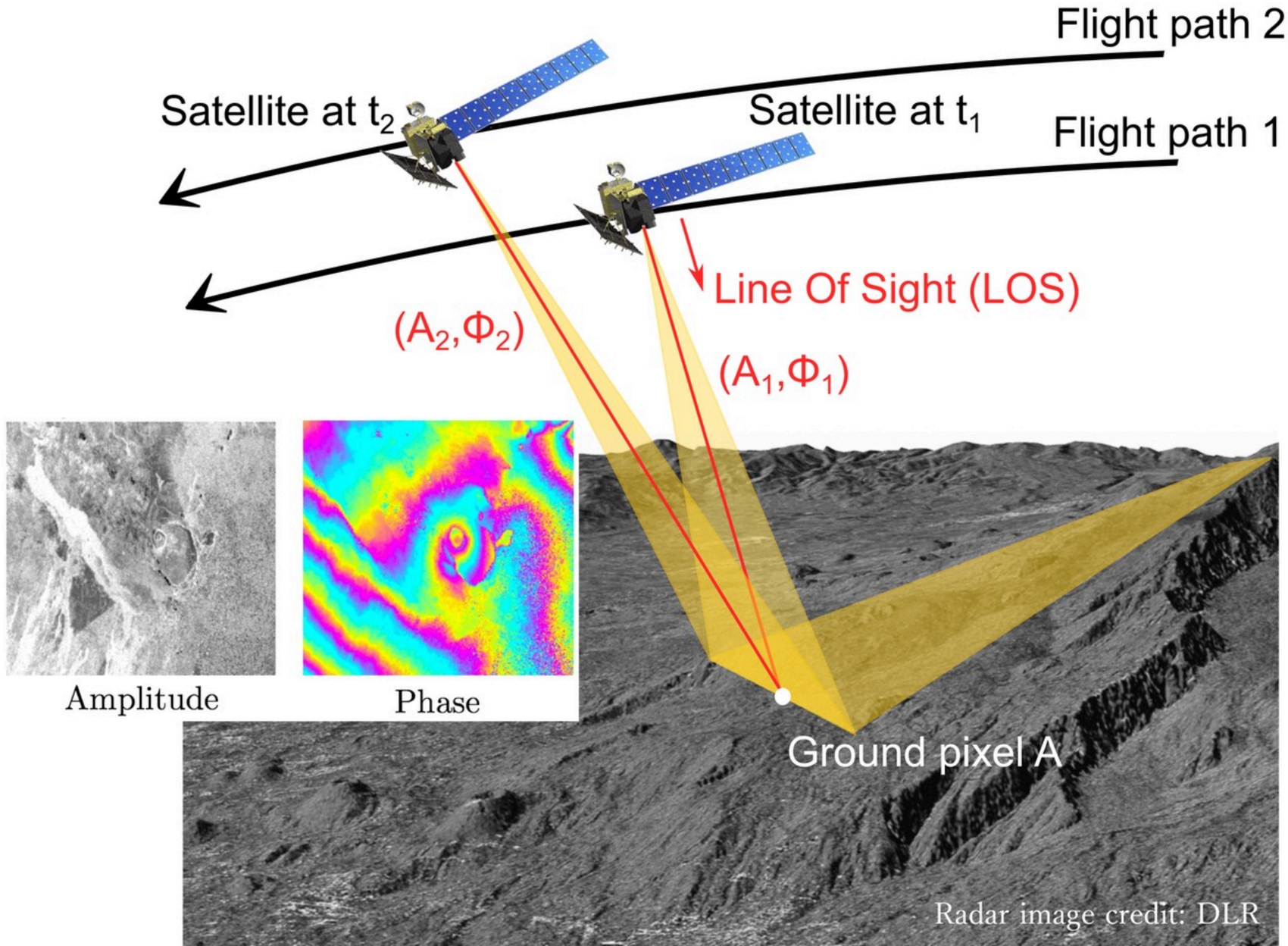

Day 4 – less flooded, not windy



# Synthetic Aperture Radar (SAR)



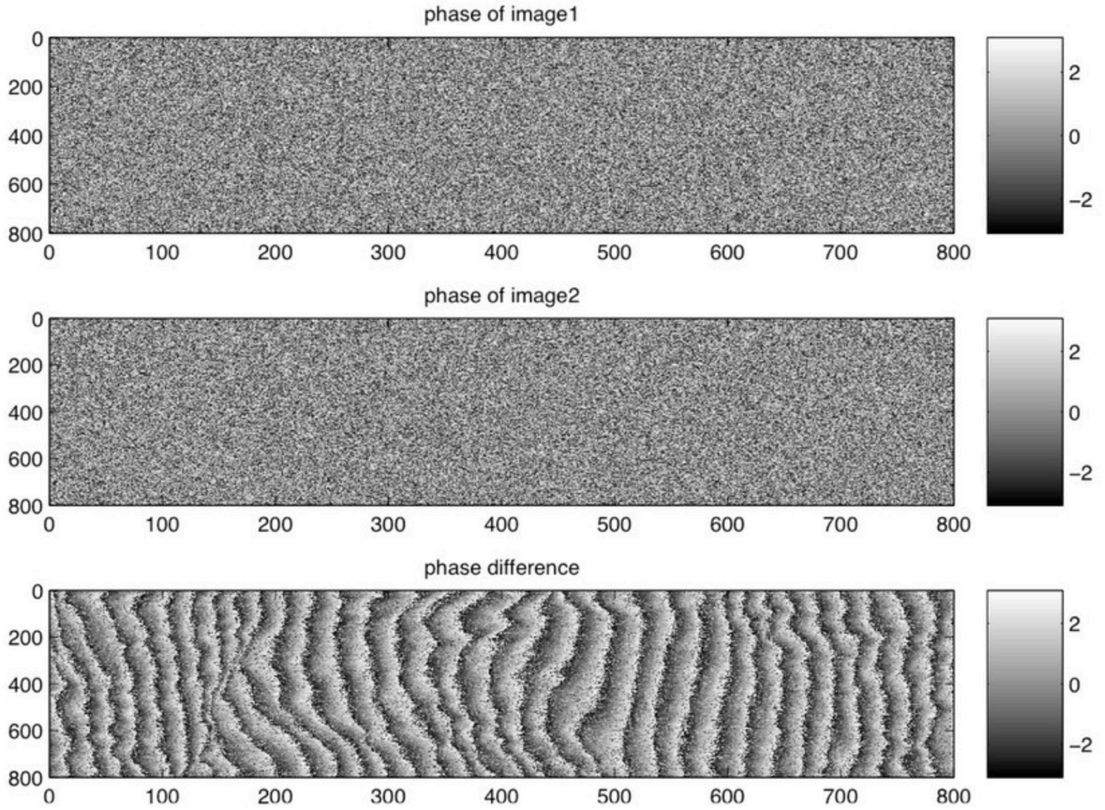
# Interferometric Synthetic Aperture Radar (InSAR)



# Interferogram Formation

$$C_2 C_1^* = A_1 A_2 e^{i(\phi_2 - \phi_1)} = R(\mathbf{x}) + iI(\mathbf{x}).$$

The phase of the interferogram is $(\phi_2 - \phi_1) = \tan^{-1}\left(\frac{I}{R}\right)$.



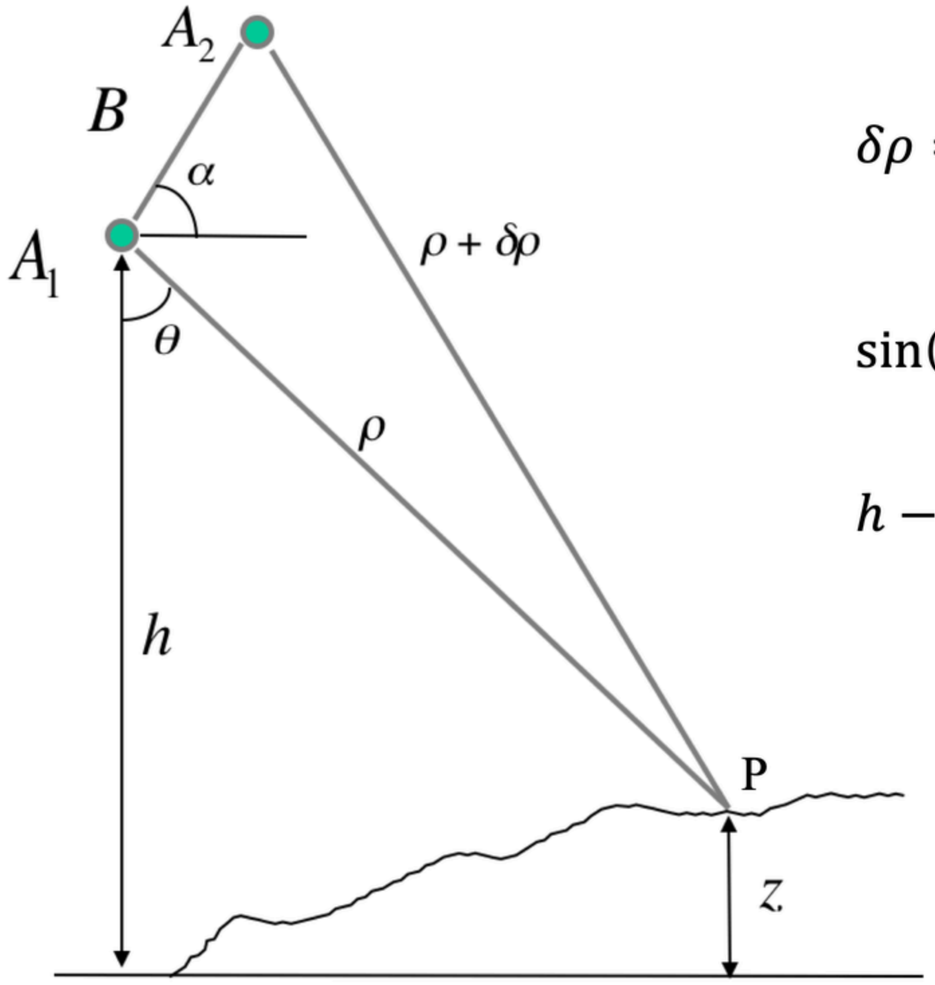
02

PART 2

# Interferometry for topography

Baselines, the critical baseline, topographic phase and DEM missions

# Interferometry for topography



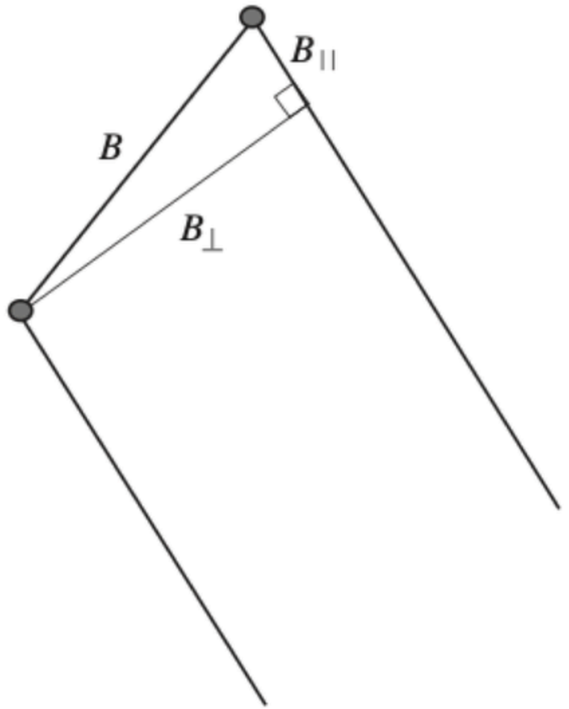

$$\delta\rho = \frac{\lambda}{2\pi p}(\phi_1 - \phi_2) = \frac{\lambda}{2\pi p}\Delta\phi$$

$$\sin(\theta - \alpha) = \frac{\rho^2 + B^2 - (\rho + \delta\rho)^2}{2\rho B}$$

$$h - z = \rho\cos\theta$$

We can measure $\Delta\phi$ (unwrapped) from radar. Given $h, B$, $\alpha$ and $\rho$, we can solve for $\theta$, which gives us $z$ at every pixel location.

# Parallel and Perpendicular Baselines




$$B_{\parallel} = B\sin(\theta - \alpha)$$ parallel-ray approximation

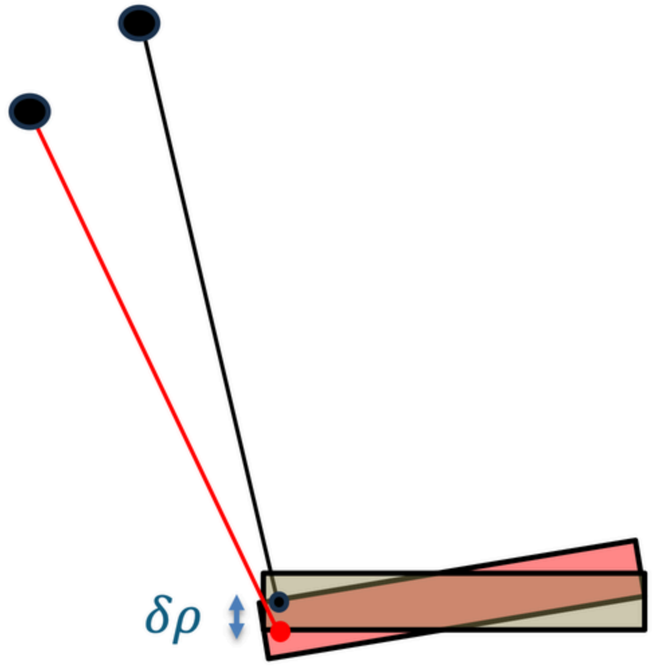
$$B_{\perp} = B\cos(\theta - \alpha)$$

# Critical Baseline

The phase change cannot be greater than $2\pi$ radians or it will be *aliased* - unrecoverable.

Recall, phase change is twice the range change (two-way travel)

$$B_c = \frac{\lambda\rho}{2R_r \cos\theta} = \frac{\lambda\rho}{C\tau}\tan\theta$$




Critical baseline for earlier satellites

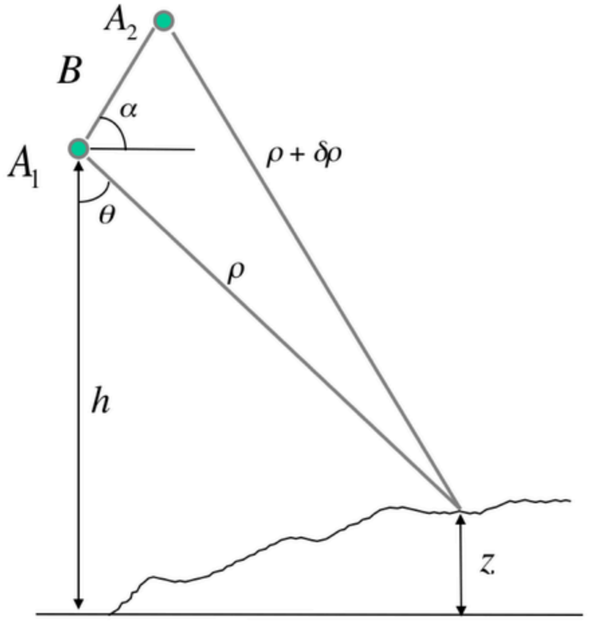
| **Look angle** | 23° | 34° | 41° |
|---|---|---|---|
| ERS/ENVISAT 16 MHz | 1.1 km | 2.0 | 2.9 |
| ALOS FBD 14 MHz | 3.6 | 6.5 | 9.6 |
| ALOS FBS 28 MHz | 7.3 | 13.1 | 18.6 |

**Values for Sentinel-1:**
Wavelength (λ): 0.055 m
Range (ρ): 800 km
Range resolution (Rr): 5 m
Incidence angle (θ): 30°

→ Critical baseline: ~ 5.1 km

# Solution for topography from interferometric phase

- After subtracting a flat earth fringe:



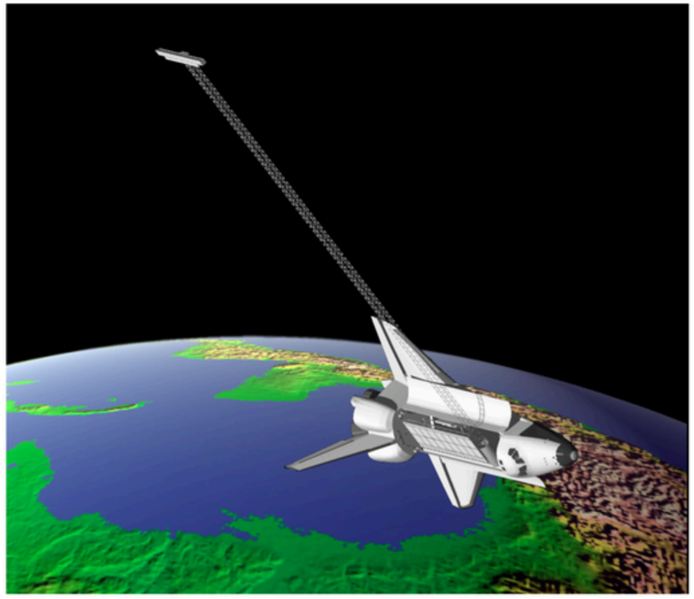

$$\phi_{topo} = -\frac{2\pi pB}{\lambda\rho}\left(\frac{1}{\tan\theta_0}\cos\alpha + \sin\alpha\right)\mathrm{dz}$$

Known!

The constant-phase contours will appear at the constant height on the imaged topography. The phase will repeat every $2\pi$ radians, so therefore each “fringe” (one phase cycle) corresponds to a constant height difference (the ambiguity height).

# Implementation options

- A single spacecraft with two displaced antennas
- Two spacecraft, each with a synthetic aperture radar, flying in formation to form the interferometer baseline.



The Shuttle Radar Topography Mission



**TanDEM-X** (**T**erraSAR-X **a**dd-o**n** for **D**igital **E**levation **M**easurement)

Technology for generation of DEM

- Optical-stereo instrumentation
- Laser profiling instruments
- Radar interferometry



DEM From USGS

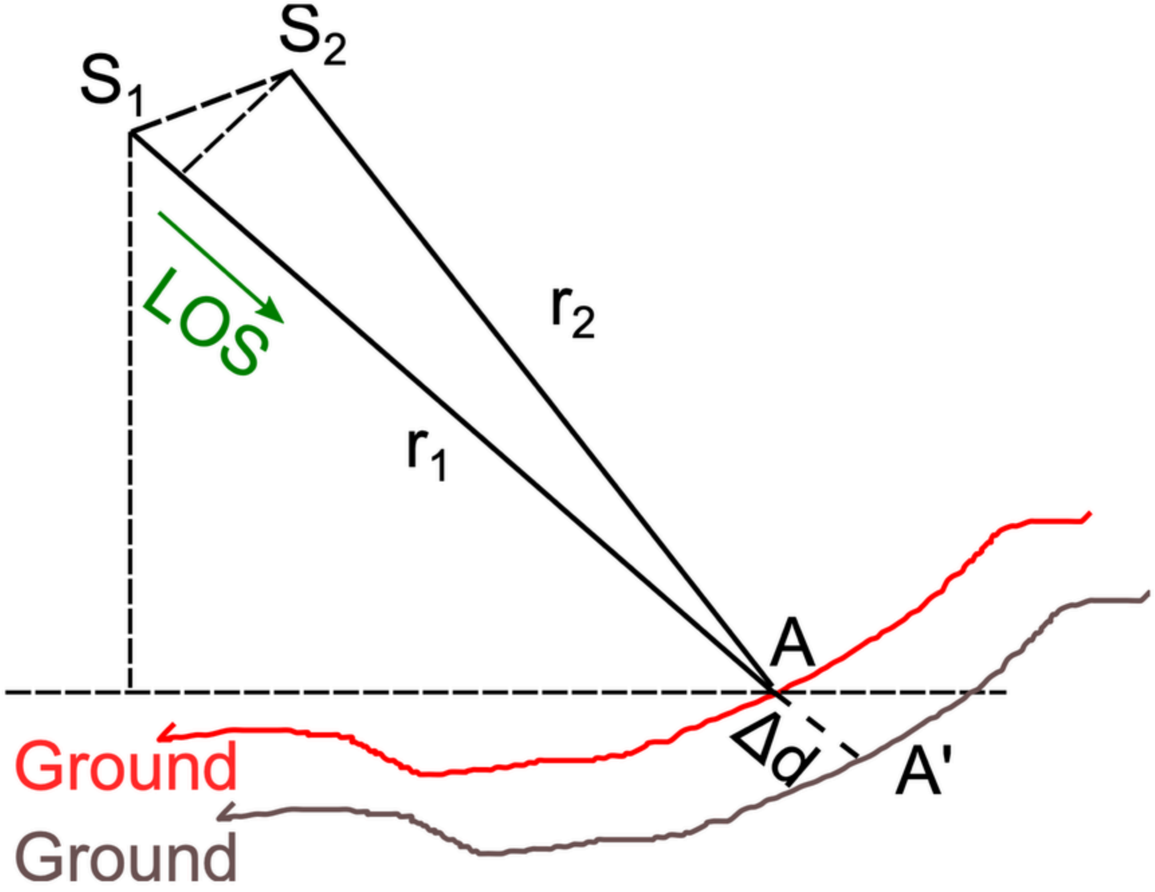
03

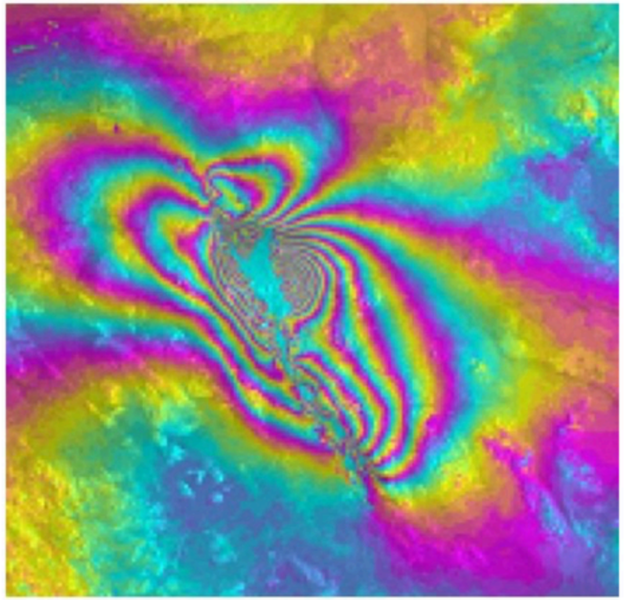
PART 3

# Interferometry for surface deformation

Deformation phase, the Landers earthquake, sensitivity and phase unwrapping

# Interferometry for surface deformation

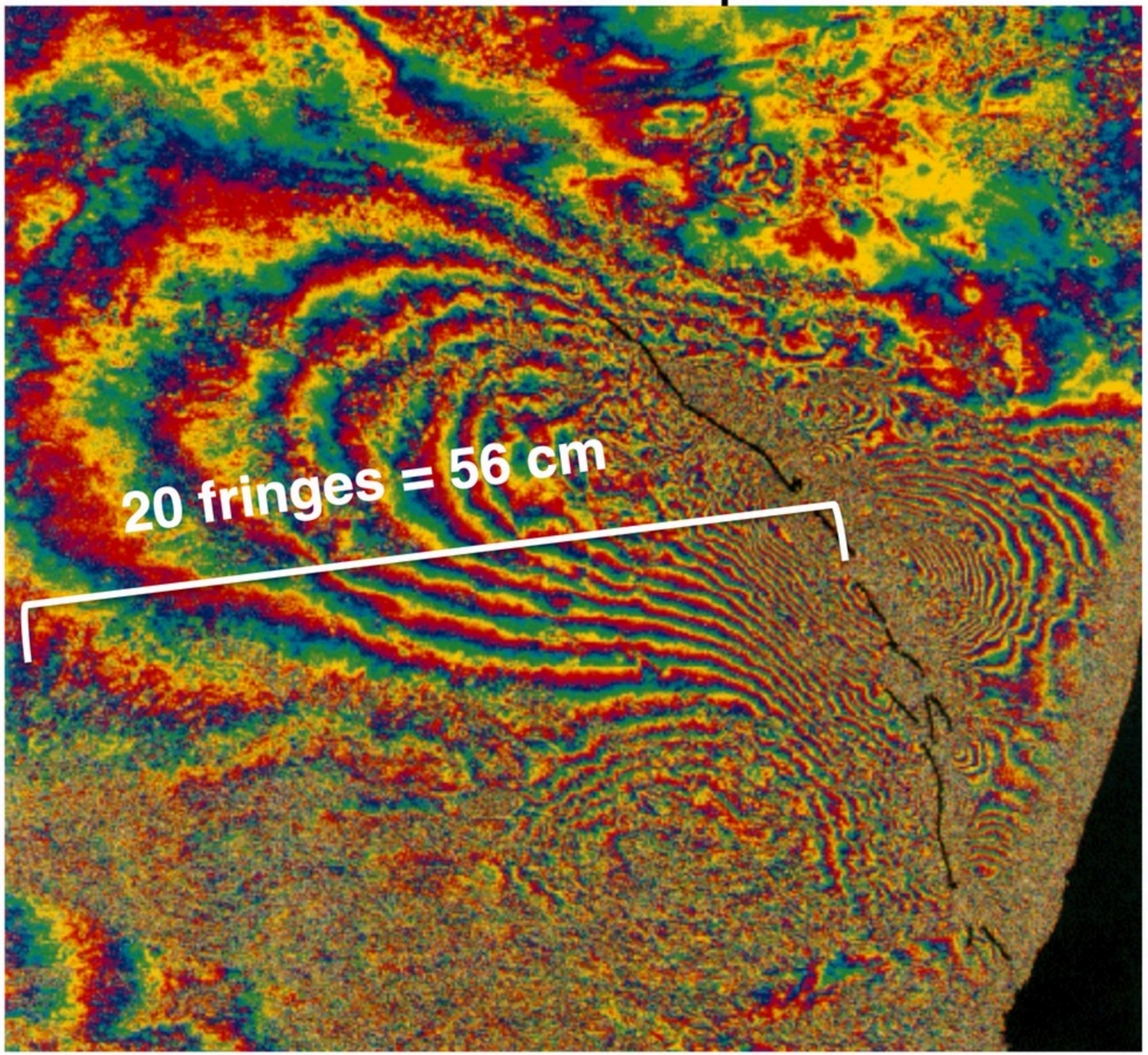


$$\phi_2 - \phi_1 = \phi_{topo} + \frac{4\pi}{\lambda}\Delta d$$

λ: radar wavelength
Δd: LOS deformation

# Interferometry for surface deformation: 7.3 Mw Landers Earthquake




June 28, 1992
ERS-1 April 24 –
August 7, 1992
$\lambda$ = 5.6 cm
90 km x 110 km area

Modified from
Massonnet et al., 1993

0 25 km

# Interferometry Sensitivities

- Topographic Sensitivity (meter level topography)

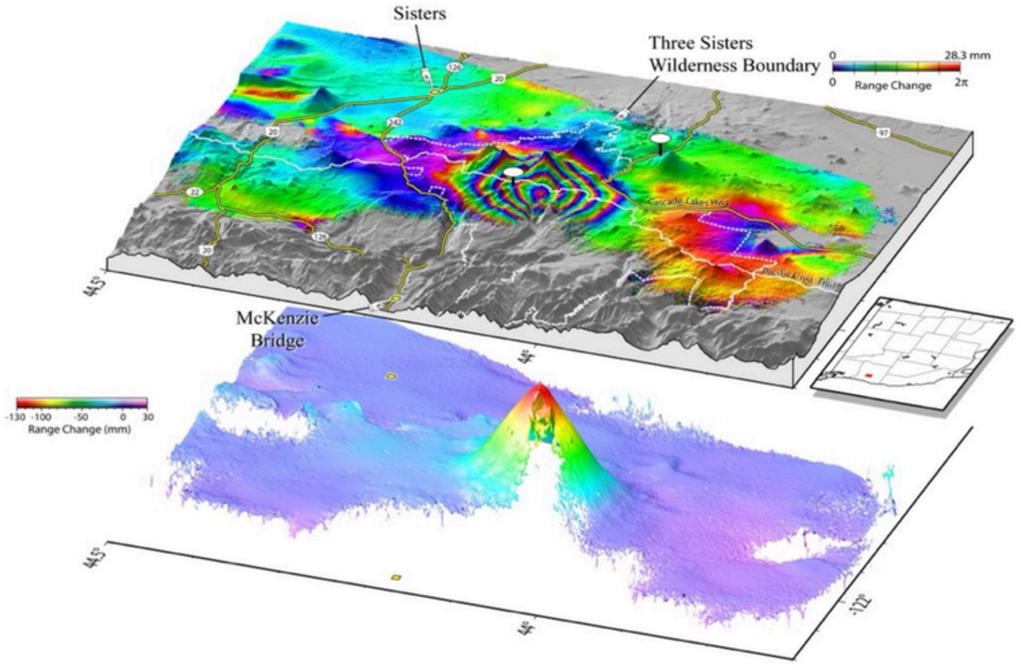
$$\frac{\partial \Delta \phi}{\partial h} = \frac{2\pi p b \cos(\theta - \alpha)}{\lambda \rho \sin \theta}$$

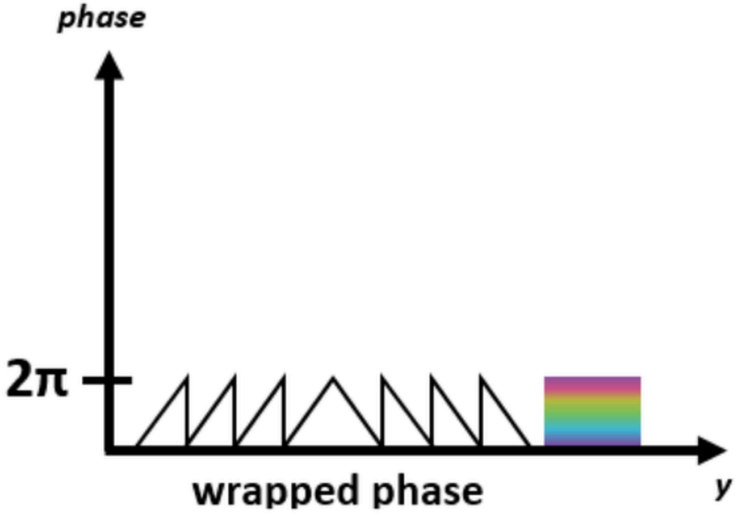
- Displacement Sensitivity (cm-mm level deformation)

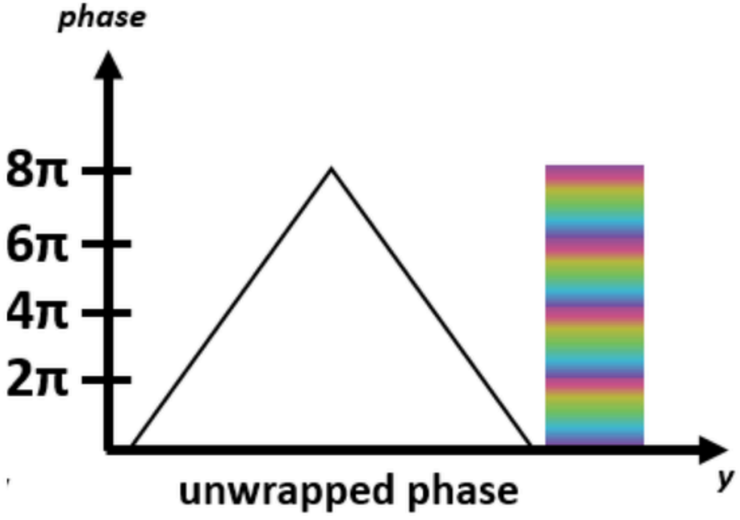
$$\frac{\partial \Delta \phi}{\partial \Delta \rho} = \frac{4\pi}{\lambda}$$

# Phase unwrapping

Credit: ESA

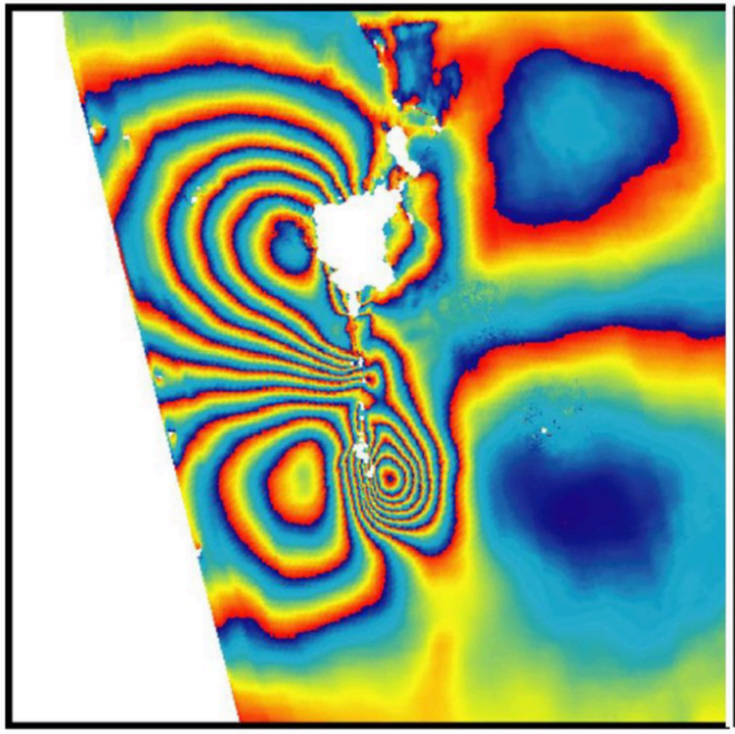


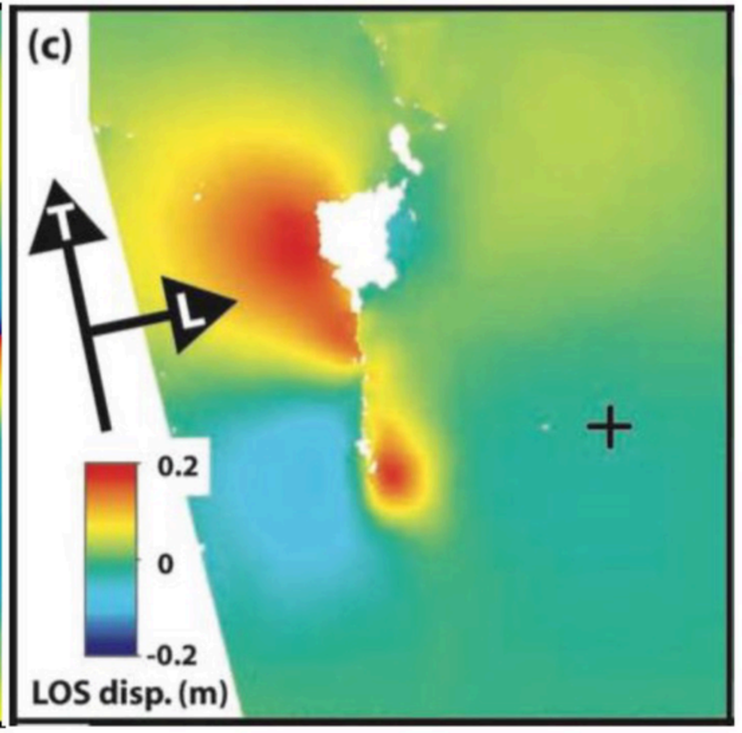



# Phase unwrapping

**Wrapped**



**Unwrapped**



04

PART 4

# Phase contributions and error sources

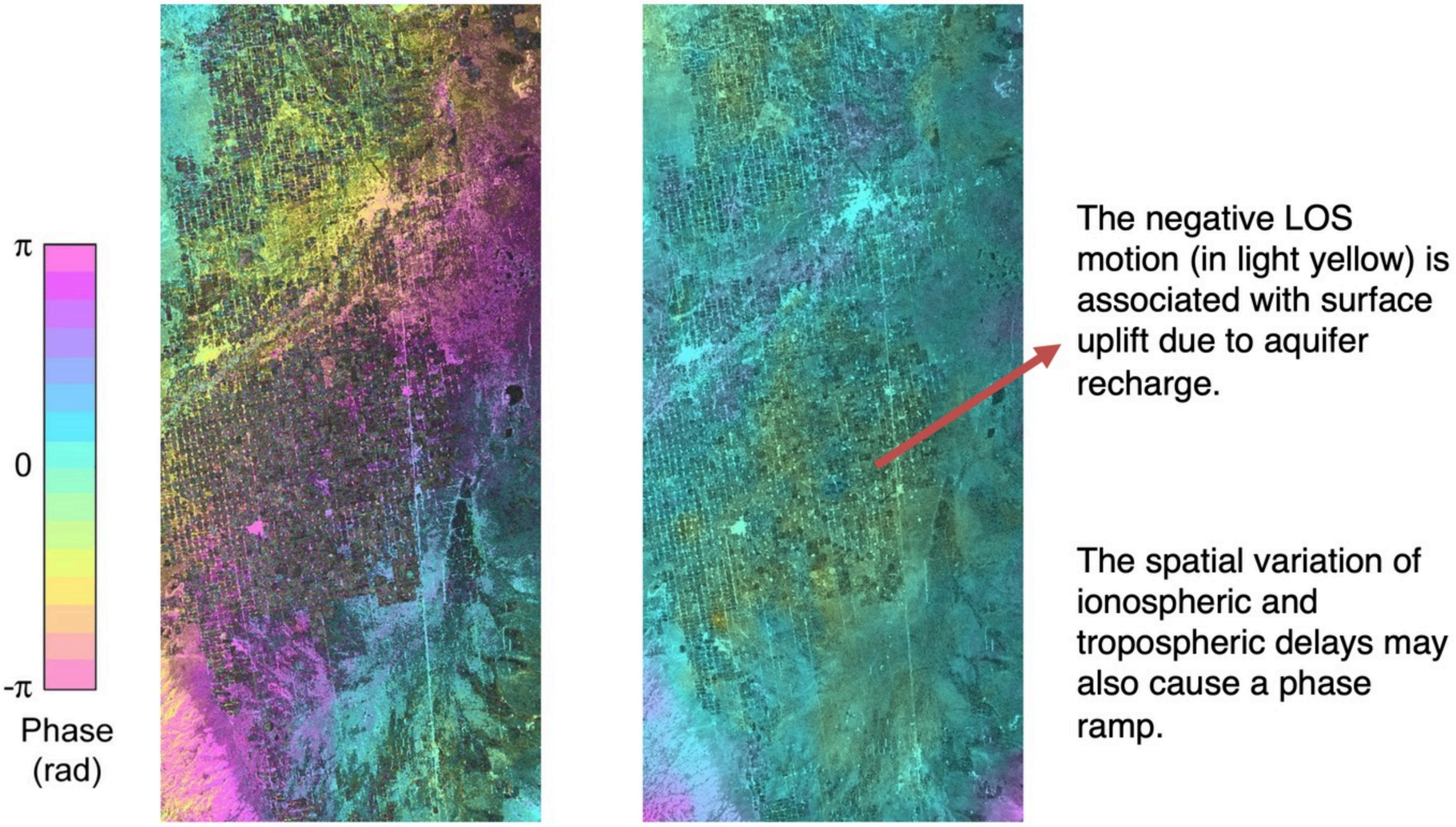
Orbits, DEMs, the ionosphere and the troposphere

# Phase contributions and error sources

$$\Delta\varphi = \frac{4\pi}{\lambda}\Delta d_{\mathrm{LOS}} + \Delta\varphi_{dem} + \Delta\varphi_{atm} + \Delta\varphi_{iono} + \Delta\varphi_{orb} + \Delta\varphi_{decor} + \Delta\varphi_{unwrp} + \Delta\varphi_{n}$$

- Orbital errors
- DEM errors
- Ionospheric and tropospheric errors
- Decorrelation and phase unwrapping errors

# Orbital errors

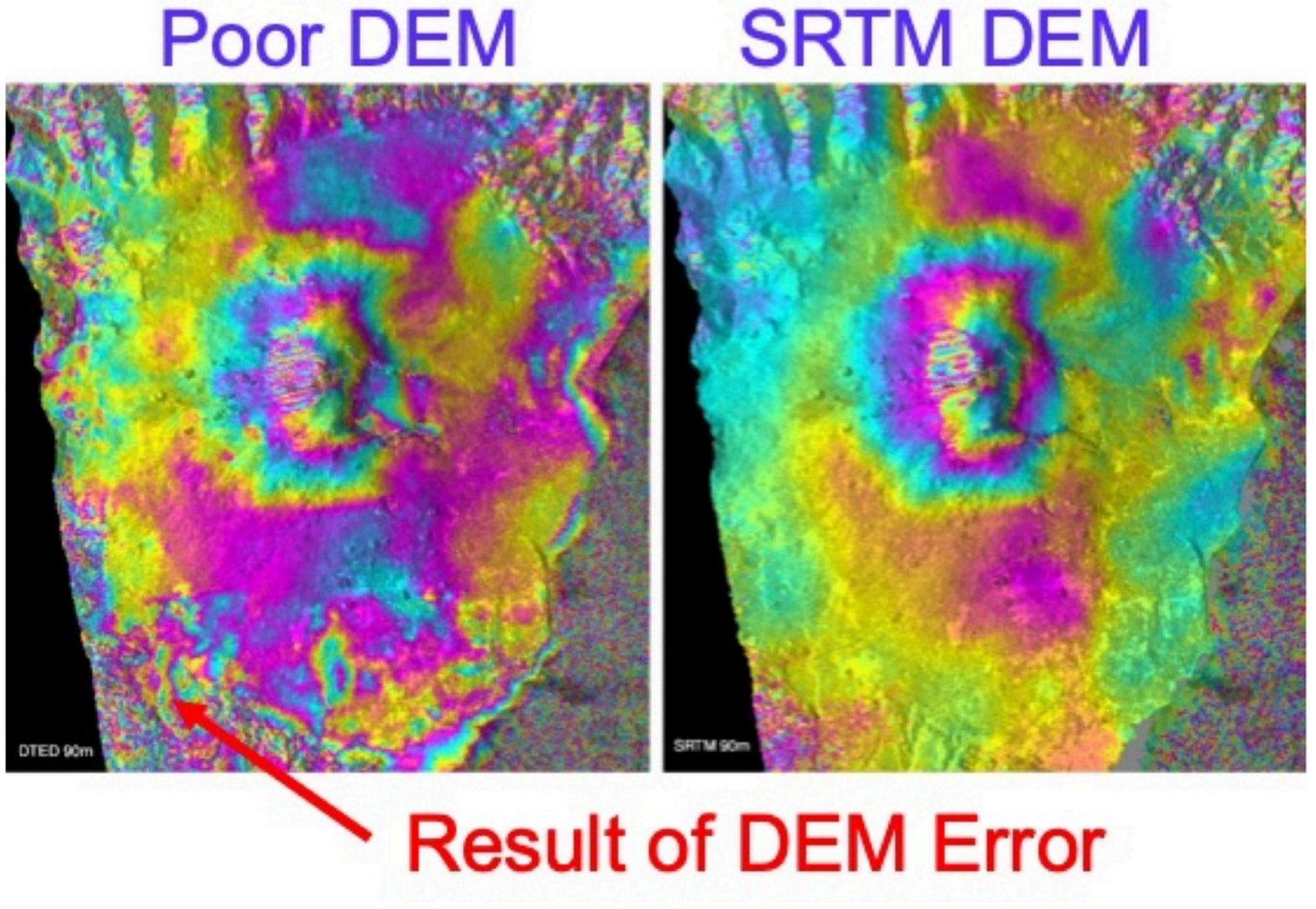



The negative LOS motion (in light yellow) is associated with surface uplift due to aquifer recharge.

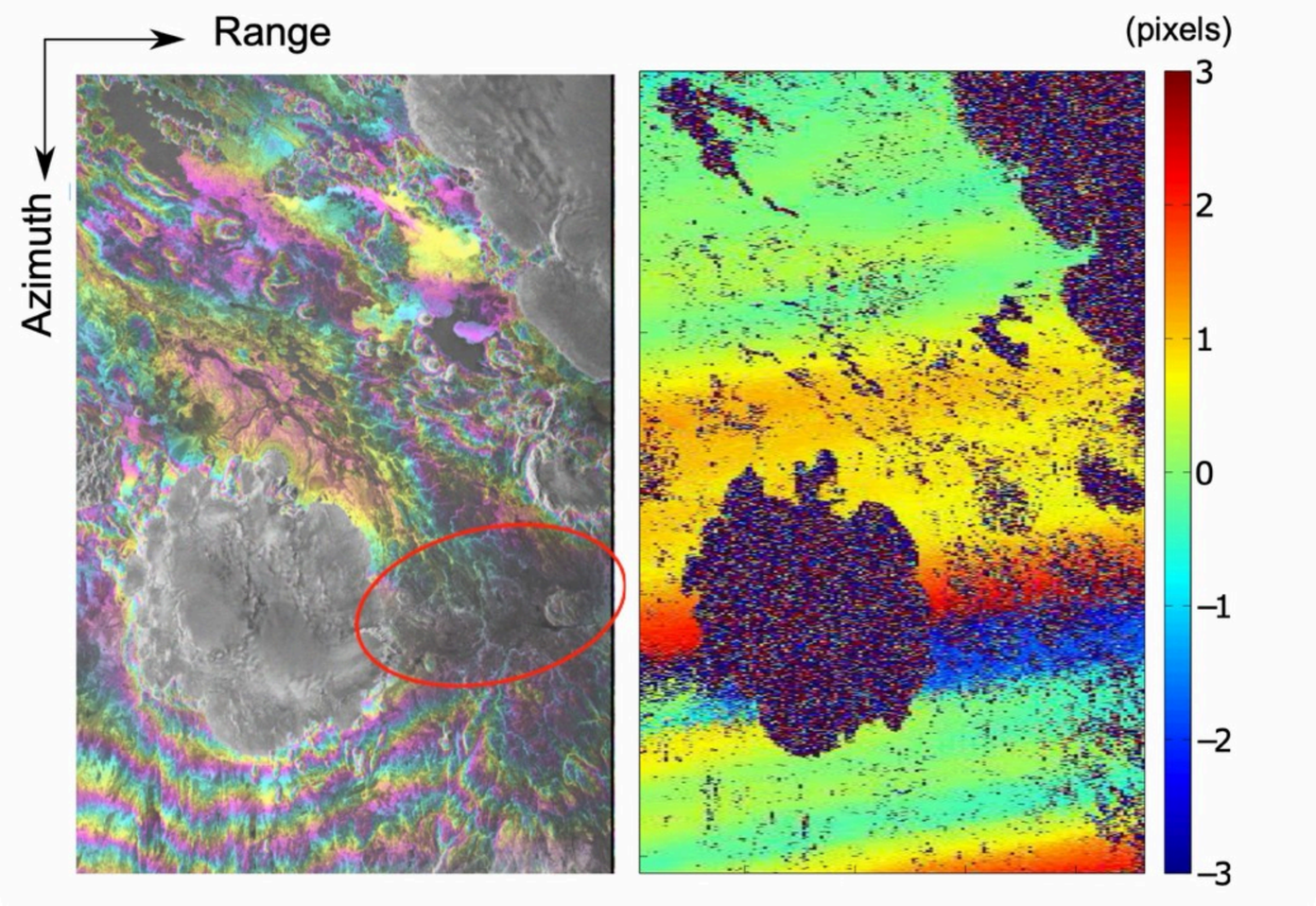
The spatial variation of ionospheric and tropospheric delays may also cause a phase ramp.

An L-band interferogram before (left) and after (right) the orbital ramp removal.

# DEM errors



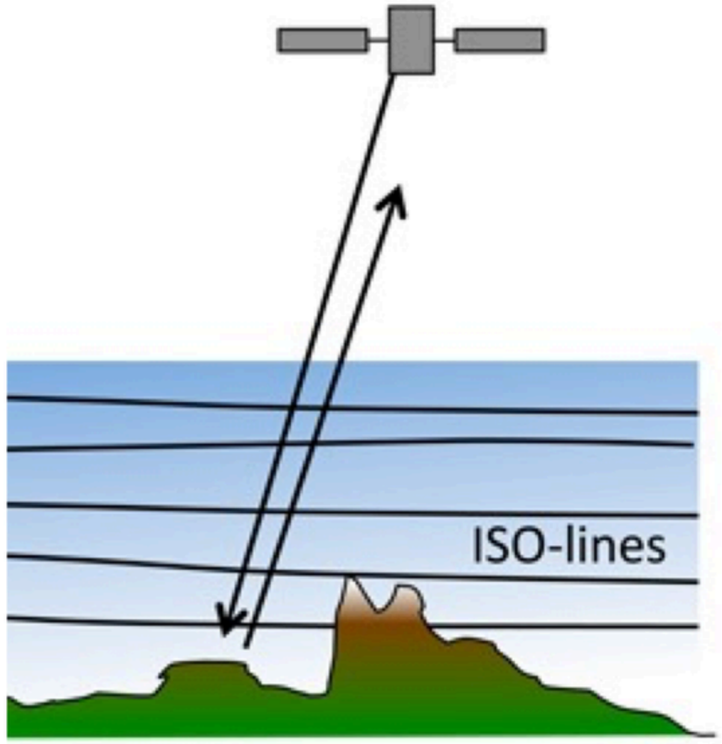
# Ionospheric artifacts



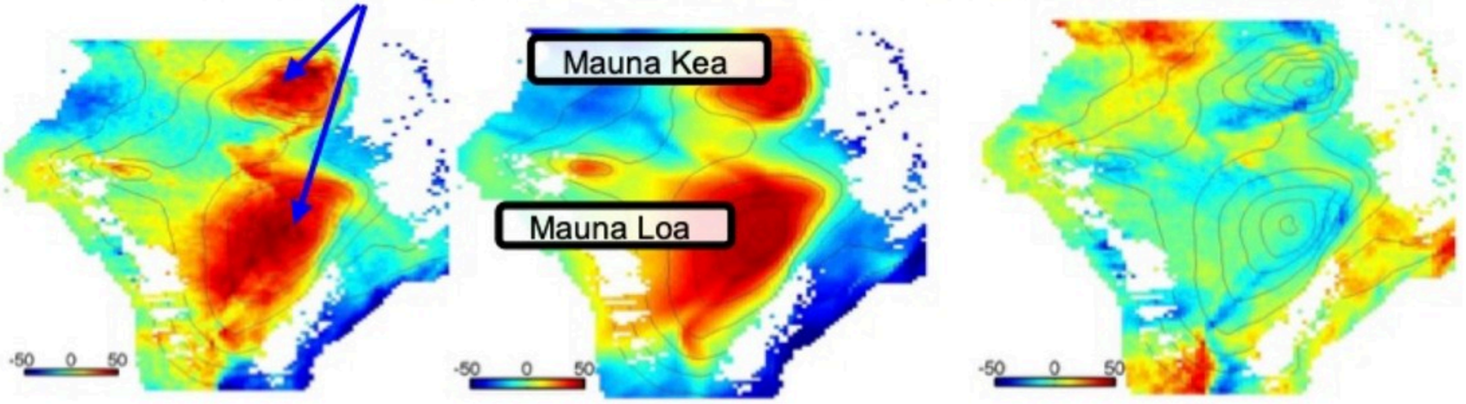

An L-band Iceland interferogram (Left) and the azimuth misregistration measured in pixels due to a local ionospheric TEC gradient.

# Tropospheric noise

- Introduced by variation in pressure, temperature, and water vapor in troposphere.
- Affect all radar wavelengths in the same way
- Can be as large as 15 cm or more
- May contain a component that is correlated with topography
- Often contain a component that is random in time.



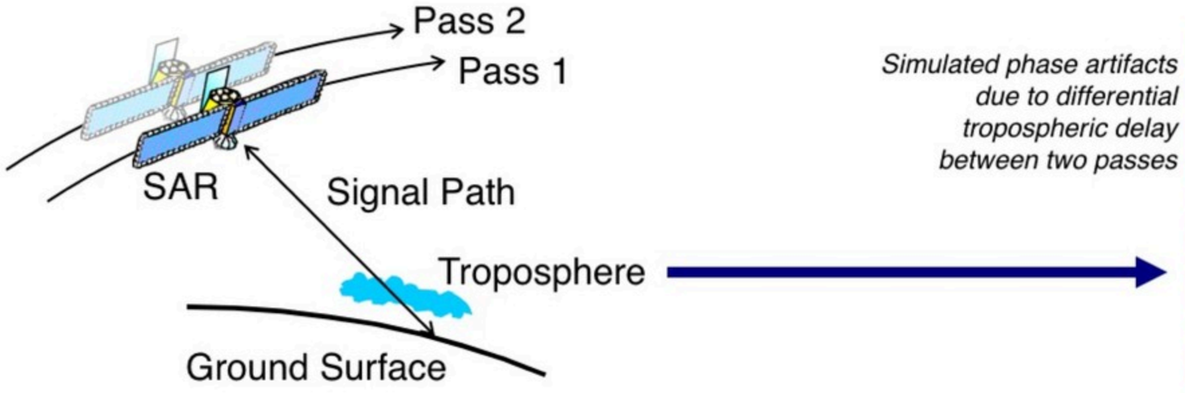
# Atmospheric phase correlated with Topography



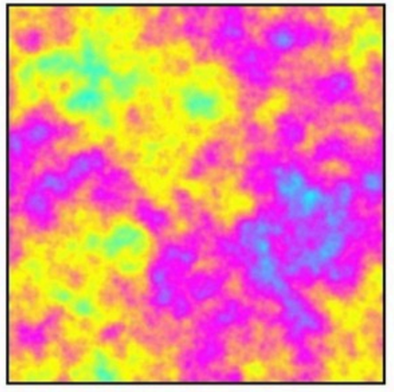

d-InSAR Observations, Big Island, Hawaii

Tropospheric Model

Phase after subtracting the Tropospheric Model




*Simulated phase artifacts due to differential tropospheric delay between two passes*



05

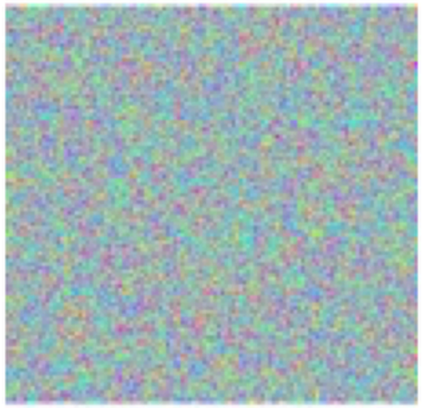
PART 5

# Correlation and wavelength

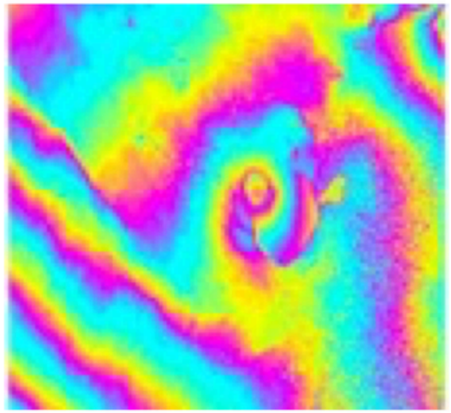
Interferogram quality, decorrelation, and X-, C-, L- and P-band compared

# Correlation

- An important measure of interferogram quality is the correlation of the two SAR images. This quantity varies from 0 to 1.



Correlation equals 0 (decorrelated), phase measurements are completely random.



High correlation, phase measurements contains high quality topographic and deformation information

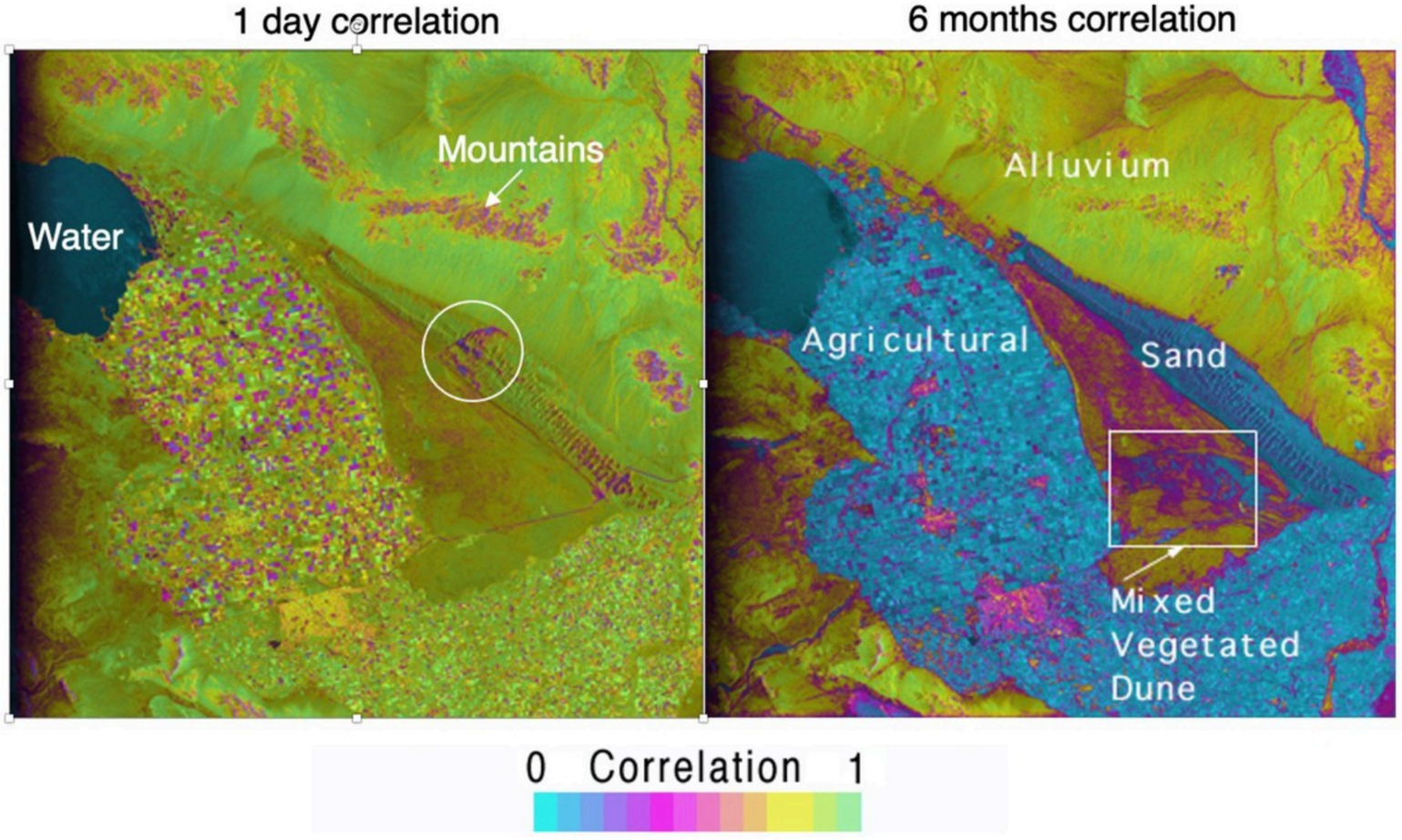
# Correlation

- We can calculate the correlation when we generate the multi-looked interferogram, according to the following definition:

$$\hat{\gamma} = \frac{\left|\langle s_1 s_2^{\star}\rangle\right|}{\sqrt{\langle s_1 s_1^{\star}\rangle\langle s_2 s_2^{\star}\rangle}}$$

$$\rightarrow \frac{\left|\sum_{mn} s_1(\rho_{mn}, x_{mn}) s_2^{\star}(\rho_{mn}, x_{mn})\right|}{\sqrt{\sum_{mn} s_1(\rho_{mn}, x_{mn}) s_1^{\star}(\rho_{mn}, x_{mn}) \sum_{mn} s_2(\rho_{mn}, x_{mn}) s_2^{\star}(\rho_{mn}, x_{mn})}}$$

# An example


1 day correlation
6 months correlation
Mountains
Water
Alluvium
Agricultural
Sand
Mixed
Vegetated
Dune
0 Correlation 1

# Total correlation

- The total correlation is the product of the individual correlation components:

$$\rho_{total} = \rho_{thermal} \cdot \rho_{spatial} \cdot \rho_{temporal}$$

where $\rho_{thermal}$ represents electronic changes inside the radar system, $\rho_{spatial}$ referred to a change in sensor location (including rotation), $\rho_{temporal}$ represents decorrelation from a change in the surface between observations.

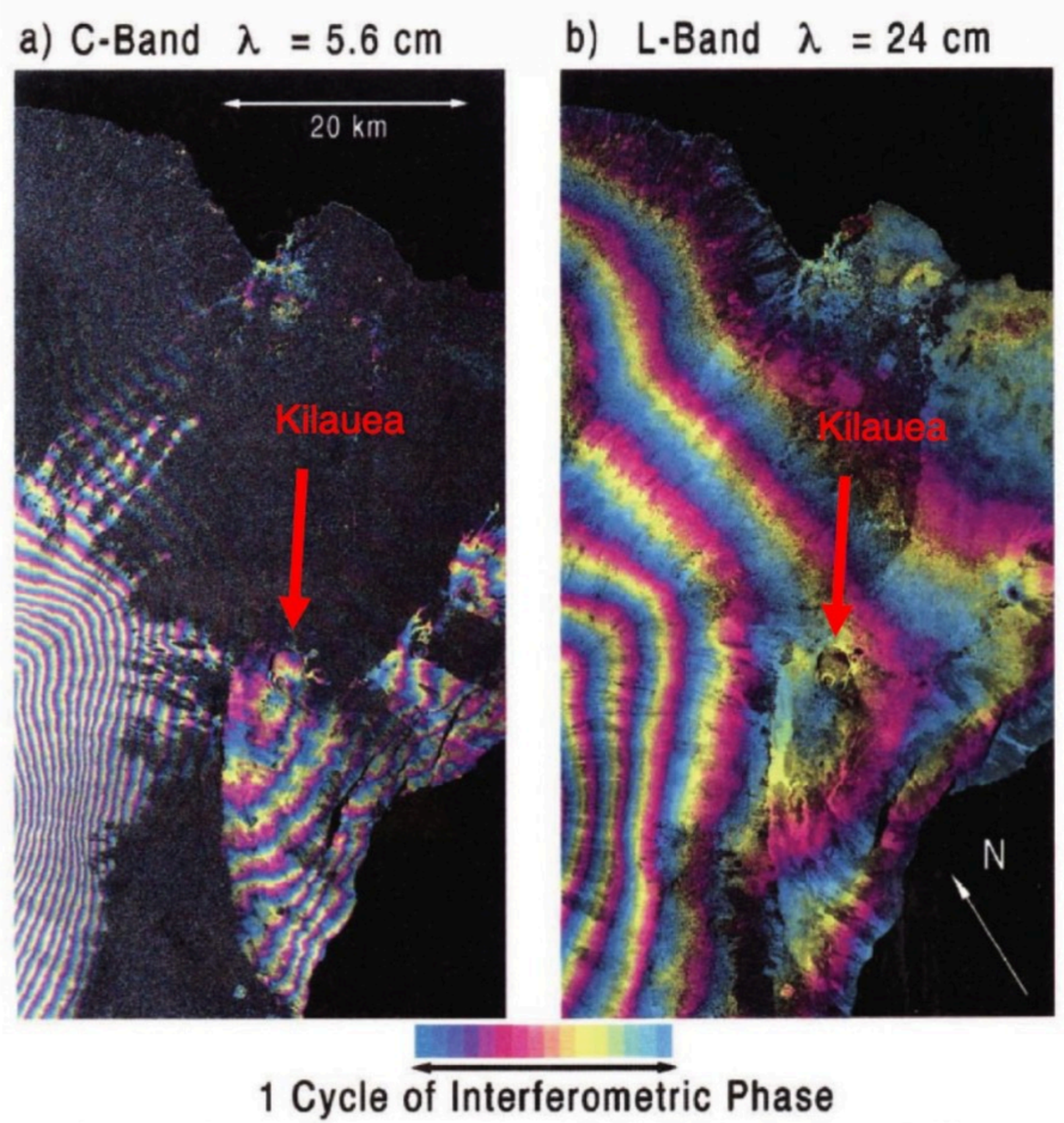


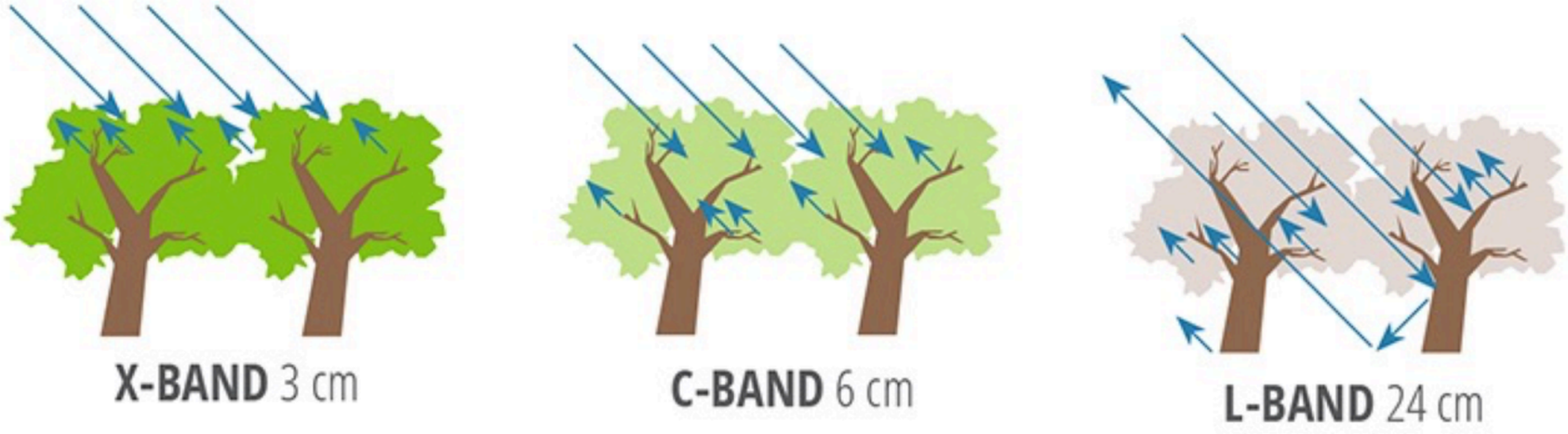

**Plate 1.** SIR-C interferograms of Kilauea, Hawaii, at (a) C band and (b) L band formed from April and October images. One cycle of interferometric phase change, or "fringe," is coded as one excursion around the color wheel. Color is modulated in brightness by the magnitude of the mean cross correlation of the images $|\langle c_1 c_2^* \rangle|$. Bright areas have high correlation and good fringe quality. Dim areas have poor correlation and fringe quality. One L band (C band) fringe corresponds to 570 m (135 m) of topographic variation.

**improved coherence**
[Rosen et al., JGR, 1996]

lower fringe rate,
easier phase unwrapping

ionospheric delay 16x worse
at L-band

# Wavelengths for different targets



# X-band

~3 cm

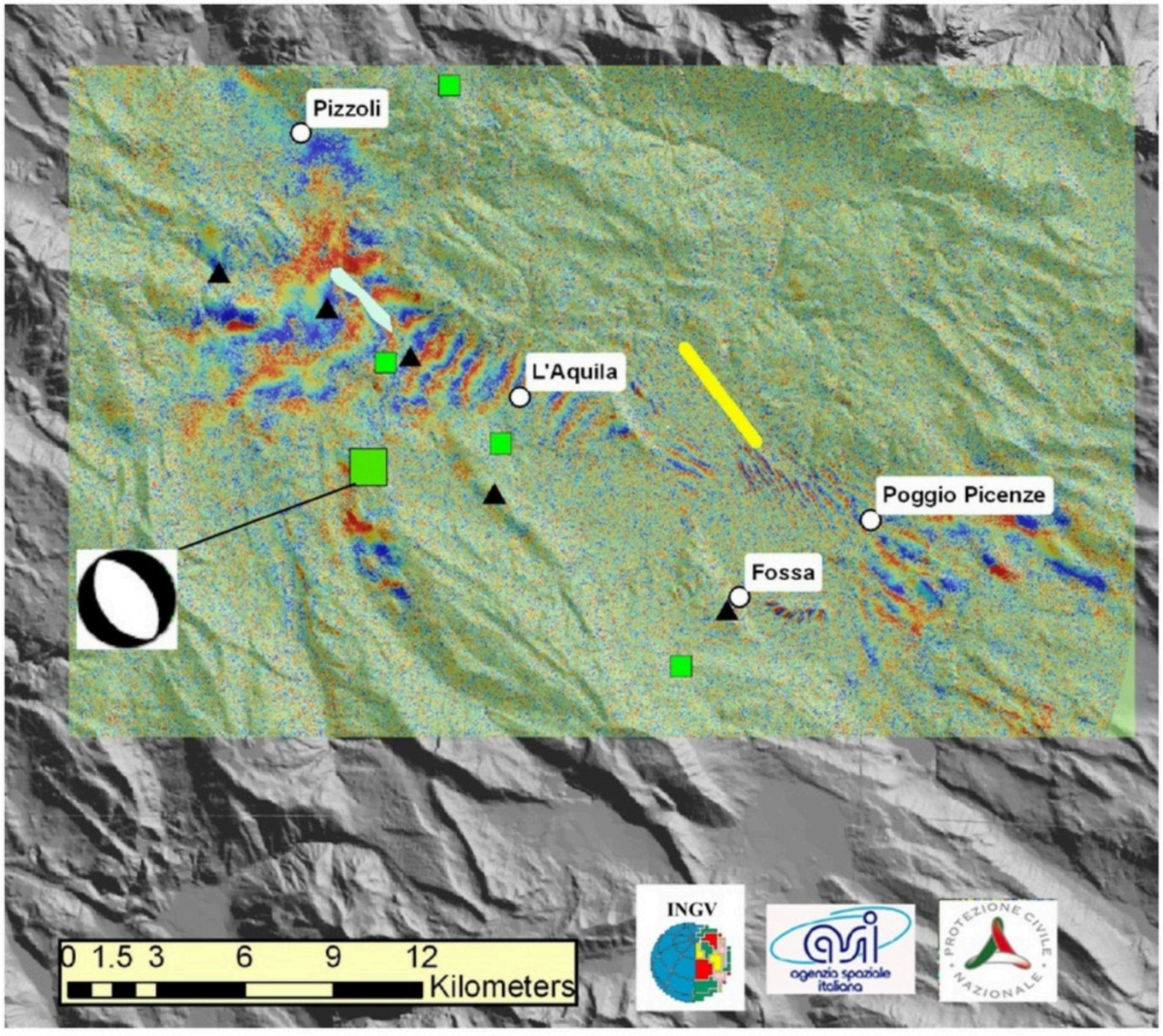
TerraSAR

Cosmo SkyMed

Iceye

Capella

On April 6, 2009 (UTC), magnitude 6.3 earthquake occurred in central Italy.

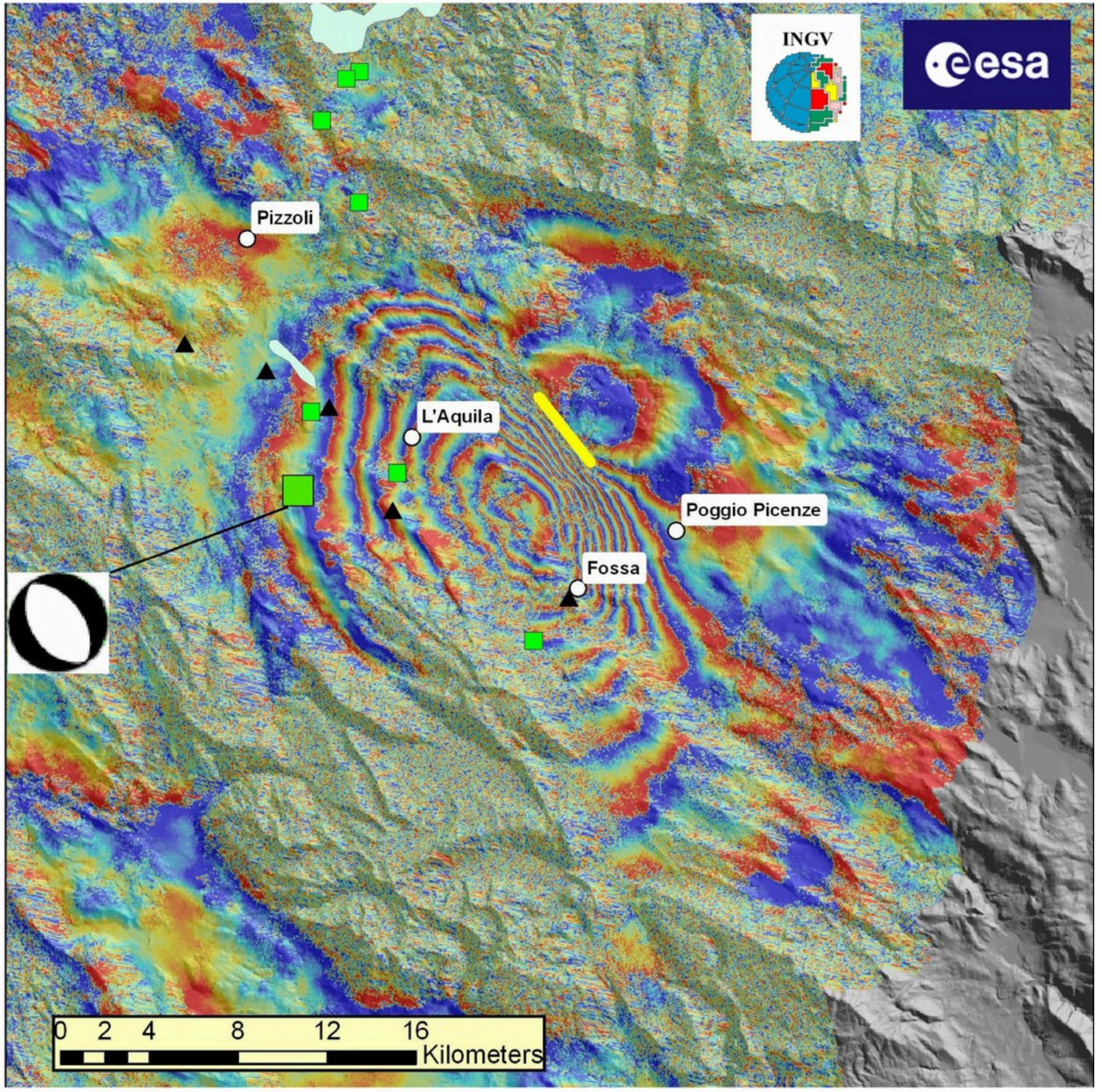



**COSMO-SkyMed** interferogram using data from 19 February 2009 and 9 April 2009. Perpendicular baseline is 480 m, and the satellite's right-looking angle is 37 degrees.

http://www.esa.int/esaEO/SEM4PJ9NJTF_index_1.html

# C-band

~6 cm
ERS-1/2
Radarsat-1/2
Envisat
Sentinel

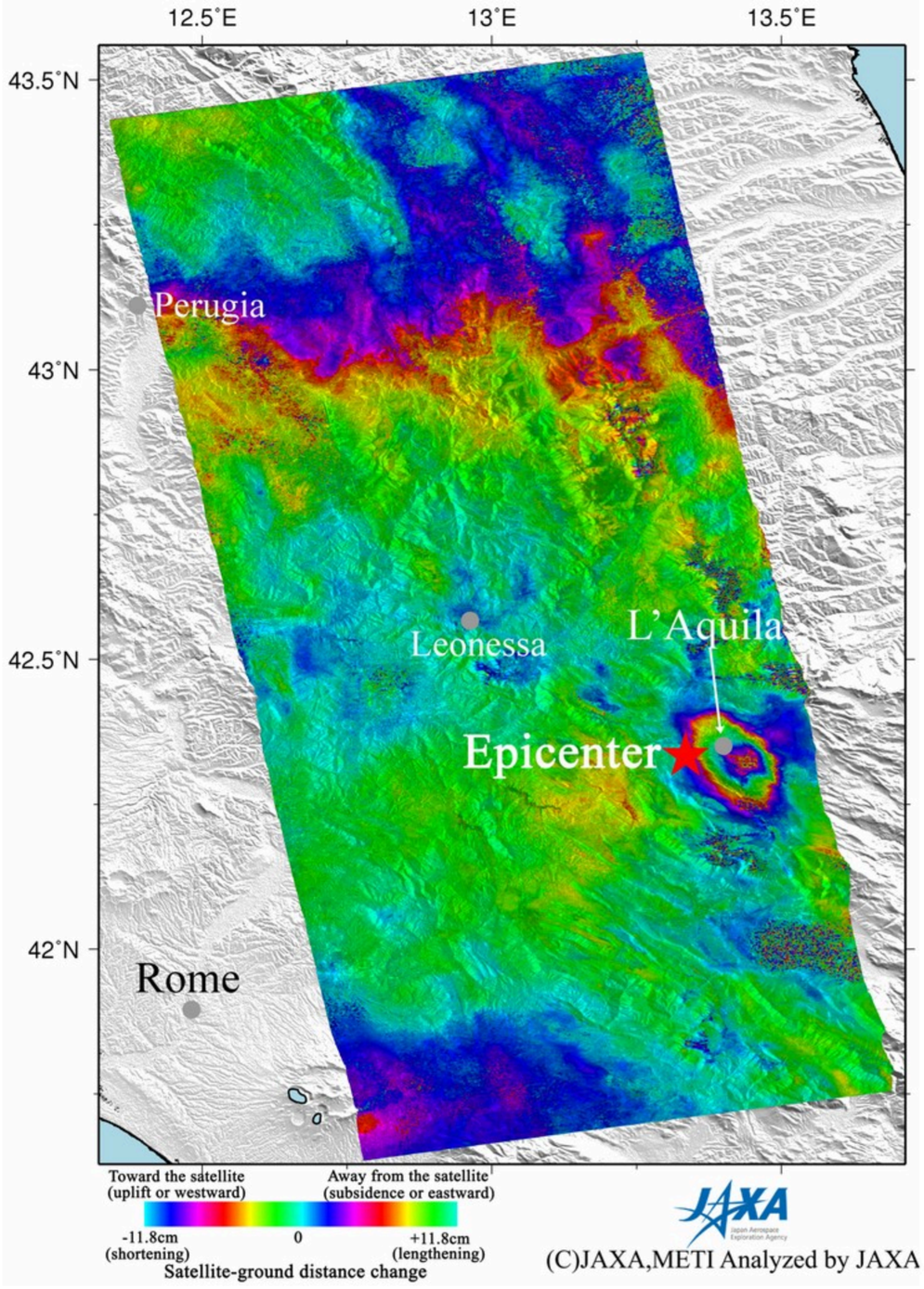



**Envisat** interferogram interpretation by Italy's Istituto Nazionale di Geofisica e Vulcanologia (INGV).

http://www.esa.int/esaEO/SEM4PJ9NJTF_index_1.html

# L-band

~24 cm
JERS-1
ALOS-1/2
UAVSAR
NISAR

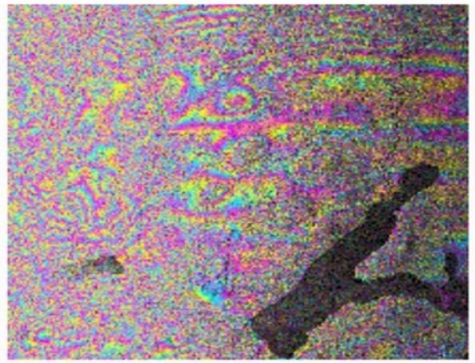



**ALOS-1**

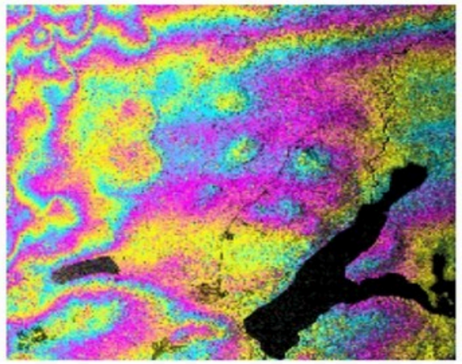
In this report, we conducted differential interferometric SAR (DInSAR) analysis to detect crustal deformation using the data acquired on April 22, 2009 and July 20, 2008

http://www.eorc.jaxa.jp/ALOS/en/img_up/dis_italia_eq_090423.htm

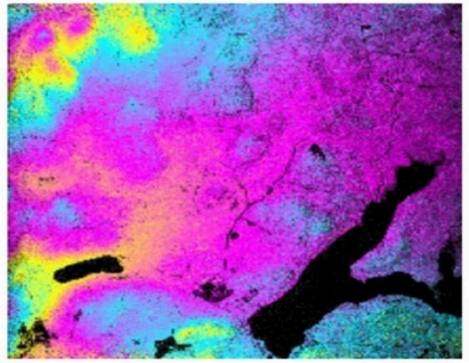
## Interferograms



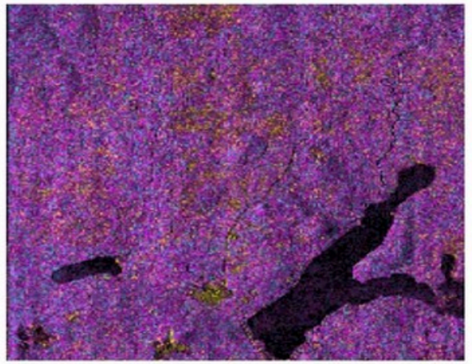
C-band

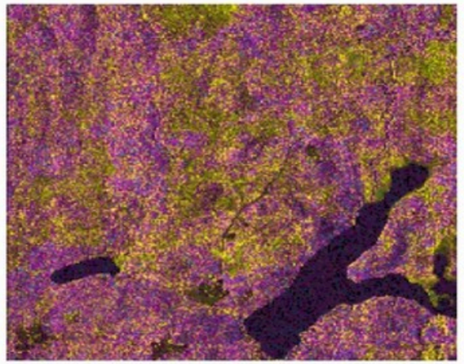


L-band

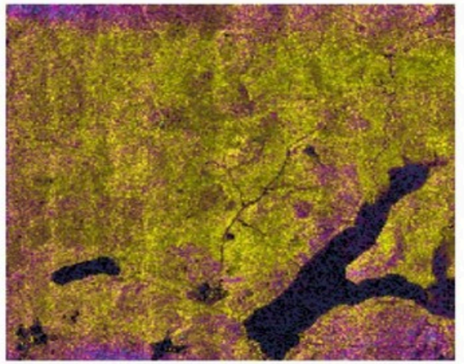


P-band





Correlation



RECAP

# Key numbers from this lecture

**$\lambda / 2$**

Line-of-sight change per fringe in repeat-pass InSAR: 2.8 cm at C-band, 11.8 cm at L-band

**56 cm**

Landers 1992: 20 C-band fringes of line-of-sight change

**$\approx$ 5.1 km**

Critical baseline for Sentinel-1 (λ = 5.5 cm, ρ = 800 km, 5 m resolution, 30°)

**15 cm+**

Possible tropospheric delay, enough to mimic real deformation

COURSE MATERIALS ON GITHUB

# Geo-SInC: the InSAR short-course repository

Repository

**github.com/parosen/Geo-SInC**

Free Jupyter notebooks and slides from the EarthScope (formerly UNAVCO) InSAR Processing and Analysis short course, 2021–2024. Today's slides are **1.3_InSAR_theory / InSARtheory.pdf**.

**PRE-COURSE**

Python, GIS with GDAL, data access and SAR theory (Lecture 1)

**DAY 1**

Geophysical modelling, InSAR theory (Lectures 2 and 3), stripmap processing with ISCE2

**DAY 2**

Interpreting coseismic interferograms; TOPS (Sentinel-1) processing

**DAY 3**

Glaciers, preparing data for modelling, atmospheric errors

**DAY 4**

Time-series theory, stacks with ARIA-tools, offset stacks

**DAY 5**

Time-series analysis with MintPy and stack processing with ISCE2

Next in the same folder: Correlation.ipynb, which goes deeper into decorrelation. To get everything: on GitHub, choose Code → Download ZIP.

# Thank you. Questions?

Lecture slides by Molly Zebker (University of California San Diego) and Ann Chen (The University of Texas at Austin), originally developed by Paul A. Rosen and Scott Hensley; EarthScope InSAR short course, github.com/parosen/Geo-SInC. Figure credits as shown on the slides, including DLR, ESA, USGS, INGV, ASI, JAXA, Massonnet et al. (1993) and Rosen et al. (1996).